

Homework #3

Markov's Inequality, Maximum Entropy, and Fano

Introduction to Information Theory 2026 · EC5203 / EC4310 / MM4450 · Prof. Heung-No Lee · GIST

How to submit. Put your solution in the `submission/` folder of your assignment repository: a single PDF (`submission/hw3.pdf` — scanned handwriting is fine if legible), or Markdown with L^AT_EX (`submission/hw3.md`). Commit and push before the deadline (23:59 KST); the last commit before the deadline is graded. If you worked as a group (max 3), list all members at the top — each member still submits in their own repository.

Questions. Light ones in the AI TA chat on the course homepage; rigorous derivations and proofs on the course GitHub Q&A (mention @claude). Before the deadline you get hints, never solutions.

Problem HW3.1 (C&T 2.21) *Markov's inequality for probabilities.* Let $p(x)$ be a probability mass function. Prove, for all $d \geq 0$, that

$$\mathbb{P}\{p(X) \leq d\} \log \frac{1}{d} \leq H(X). \quad (2.175)$$

Problem HW3.2 (C&T 2.30) *Maximum entropy.* Find the probability mass function $p(x)$ that maximizes the entropy $H(X)$ of a nonnegative integer-valued random variable X subject to the constraint

$$EX = \sum_{n=0}^{\infty} n p(n) = A$$

for a fixed value $A > 0$. Evaluate this maximum $H(X)$.

Problem HW3.3 (C&T 2.32) *Fano.* We are given the following joint distribution on (X, Y) :

$X \backslash Y$	a	b	c
1	$\frac{1}{6}$	$\frac{1}{12}$	$\frac{1}{12}$
2	$\frac{1}{12}$	$\frac{1}{6}$	$\frac{1}{12}$
3	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{1}{6}$

Let $\hat{X}(Y)$ be an estimator for X (based on Y) and let $P_e = \mathbb{P}\{\hat{X}(Y) \neq X\}$.

- Find the minimum probability of error estimator $\hat{X}(Y)$ and the associated P_e .
- Evaluate Fano's inequality for this problem and compare.

Problem HW3.4 (C&T 2.33) *Fano's inequality.* Let $\mathbb{P}(X = i) = p_i$, $i = 1, 2, \dots, m$, and let $p_1 \geq p_2 \geq \dots \geq p_m$. The minimal probability of error predictor of X is $\hat{X} = 1$, with resulting probability of error $P_e = 1 - p_1$. Maximize $H(\mathbf{p})$ subject to the constraint $1 - p_1 = P_e$ to find a bound on P_e in terms of H . This is Fano's inequality in the absence of conditioning.

Problem HW3.5 (C&T 3.1) *Markov's inequality and Chebyshev's inequality.*

- (a) (*Markov's inequality*) For any nonnegative random variable X and any $t > 0$, show that

$$P\{X \geq t\} \leq \frac{EX}{t}. \quad (3.31)$$

Exhibit a random variable that achieves this inequality with equality.

- (b) (*Chebyshev's inequality*) Let Y be a random variable with mean μ and variance σ^2 . By letting $X = (Y - \mu)^2$, show that for any $\epsilon > 0$,

$$P\{|Y - \mu| > \epsilon\} \leq \frac{\sigma^2}{\epsilon^2}. \quad (3.32)$$

- (c) (*Weak law of large numbers*) Let $Z_1, Z_2, \dots, Z_n$ be a sequence of i.i.d. random variables with mean μ and variance σ^2 . Let $\bar{Z}_n = \frac{1}{n} \sum_{i=1}^n Z_i$ be the sample mean. Show that

$$P\{|\bar{Z}_n - \mu| > \epsilon\} \leq \frac{\sigma^2}{n\epsilon^2}. \quad (3.33)$$

Thus, $P\{|\bar{Z}_n - \mu| > \epsilon\} \rightarrow 0$ as $n \rightarrow \infty$. This is known as the *weak law of large numbers*.

Problem HW3.6 (C&T 3.2) *AEP and mutual information.* Let (X_i, Y_i) be i.i.d. $\sim p(x, y)$. We form the log likelihood ratio of the hypothesis that X and Y are independent vs. the hypothesis that X and Y are dependent. What is the limit of

$$\frac{1}{n} \log \frac{p(X^n)p(Y^n)}{p(X^n, Y^n)}?$$

Problem HW3.7 (C&T 3.4) *AEP.* Let X_i be iid $\sim p(x)$, $x \in \{1, 2, \dots, m\}$. Let $\mu = EX$ and $H = -\sum p(x) \log p(x)$. Let $A^n = \{x^n \in \mathcal{X}^n : |-\frac{1}{n} \log p(x^n) - H| \leq \epsilon\}$. Let $B^n = \{x^n \in \mathcal{X}^n : |\frac{1}{n} \sum_{i=1}^n X_i - \mu| \leq \epsilon\}$.

- (a) Does $P\{X^n \in A^n\} \rightarrow 1$?
 (b) Does $P\{X^n \in A^n \cap B^n\} \rightarrow 1$?

Problem HW3.8 (C&T 3.10) *Random box size.* An n -dimensional rectangular box with sides $X_1, X_2, X_3, \dots, X_n$ is to be constructed. The volume is $V_n = \prod_{i=1}^n X_i$. The edge length l of an n -cube with the same volume as the random box is $l = V_n^{1/n}$. Let $X_1, X_2, \dots$ be i.i.d. uniform random variables over the unit interval $[0, 1]$. Find $\lim_{n \rightarrow \infty} V_n^{1/n}$ and compare to $(EV_n)^{\frac{1}{n}}$. Clearly, the expected edge length does not capture the idea of the volume of the box. The geometric mean, rather than the arithmetic mean, characterizes the behavior of products.

Problem HW3.9 (C&T 3.13) *Calculation of typical set.* To clarify the notion of a typical set $A_\epsilon^{(n)}$ and the smallest set of high probability $B_\delta^{(n)}$, we will calculate the set for a simple example. Consider a sequence of i.i.d. binary random variables, $X_1, X_2, \dots, X_n$, where the probability that $X_i = 1$ is 0.6 (and therefore the probability that $X_i = 0$ is 0.4).

- (a) Calculate $H(X)$.
 (b) With $n = 25$ and $\epsilon = 0.1$, which sequences fall in the typical set $A_\epsilon^{(n)}$? What is the probability of the typical set? How many elements are there in the typical set? (This involves computation of a table of probabilities for sequences with k 1's, $0 \leq k \leq 25$, and finding those sequences that are in the typical set.)

- (c) How many elements are there in the smallest set that has probability 0.9?
- (d) How many elements are there in the intersection of the sets in parts (b) and (c)? What is the probability of this intersection?

Note: the following table has incorrect entries. You should produce this table first on your own.

k	$\binom{n}{k}$	$\binom{n}{k} p^k (1-p)^{n-k}$	$-\frac{1}{n} \log p(x^n)$
0	1	0.000000	1.321928
1	25	0.000000	1.298530
2	300	0.000000	1.275131
3	2300	0.000001	1.251733
4	12650	0.000007	1.228334
5	53130	0.000054	1.204936
6	177100	0.000227	1.181537
7	480700	0.001205	1.158139
8	1081575	0.003121	1.134740
9	2042975	0.013169	1.111342
10	3268760	0.021222	1.087943
11	4457400	0.077801	1.064545
12	5200300	0.075967	1.041146
13	5200300	0.267718	1.017748
14	4457400	0.146507	0.994349
15	3268760	0.575383	0.970951
16	2042975	0.151086	0.947552
17	1081575	0.846448	0.924154
18	480700	0.079986	0.900755
19	177100	0.970638	0.877357
20	53130	0.019891	0.853958
21	12650	0.997633	0.830560
22	2300	0.001937	0.807161
23	300	0.999950	0.783763
24	25	0.000047	0.760364
25	1	0.000003	0.736966

Problem HW3.10 [Challenge, not required] L^2 and in-probability convergence of the sample mean. Let $X_1, \dots, X_n$ be uncorrelated random variables with $\mathbb{E}\{X_i\} = \mu$ and $\text{Var}(X_i) \leq C < \infty$. Prove that $(X_1 + \dots + X_n)/n \Rightarrow \mu$ in L^2 as well as in probability. (Hint: prove L^2 first and use Chebyshev's inequality.)

Problem HW3.11 [Challenge, not required] Fano lower bound on BPSK error probability. Find a tight lower bound on the probability of error for BPSK signaling over the AWGN channel. (Hint: Fano's inequality.)

- (a) Obtain the bound expression.
- (b) Numerically evaluate your expression.
- (c) Draw the bound on P_e as a function of E_b/N_0 , from -1 to 5 dB SNR in steps of 0.5 dB.